



भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

CS6903: Network Security

(Module 2: Principles of Cryptography)

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Acknowledgement

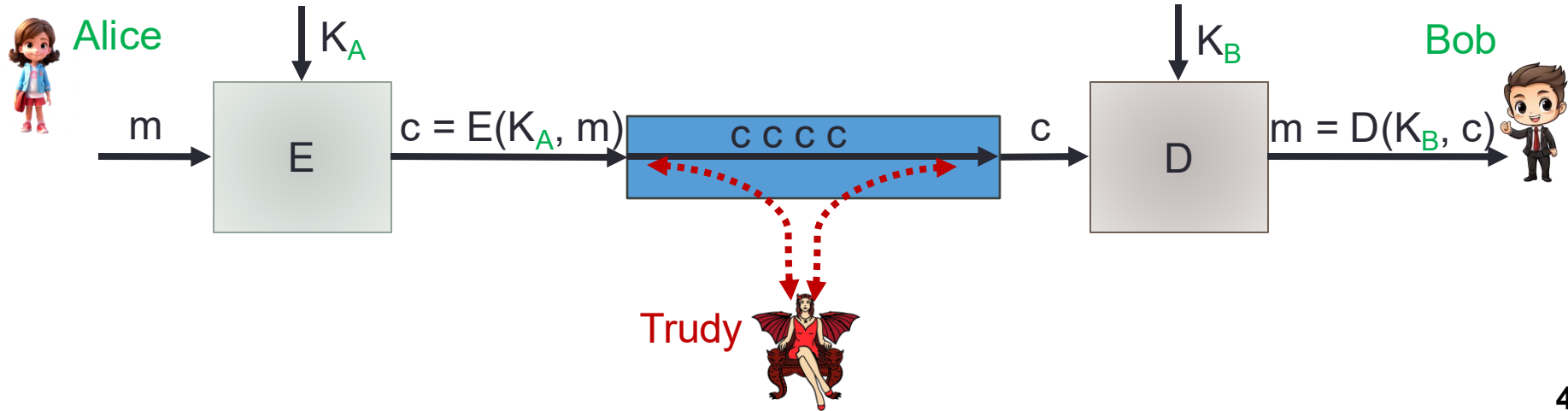
- ❑ Instructors from previous course floating
 - Prof. Bheemarjuna Reddy Tamma, Prof. Kotaro Kataoka, Prof. Antony Franklin
- ❑ Professors from other Universities/Insitutes
 - Sandeep K. Shukla, Stefan Savage, Dan Boneh, Kirill Levchenko, Alex Gantman, Deian Stefan, Nadia Heninger, Alex Dent, Vitaly Shamtikov, Robert Turner
- ❑ Other internet sources

What is Cryptography?

- ❑ Art of writing or solving (secret) code
- ❑ Modern cryptography provides
 - Confidentiality, Integrity (including authentication and non-repudiation), privacy, etc.
 - **But not availability**
- ❑ Fantastic tool to protect information in a computer systems
- ❑ Cryptography is not:
 - The solution to all security problems
 - Reliable unless implemented and used properly
- ❑ We focus primarily on using it as a (black box) tool for secure communication

Cipher: Building Block

- ❑ An algorithm to transform plaintext (m) to/from ciphertext (c)
- ❑ Include two function
 - Encryption/Encipher (E) and Decryption/Decipher (D)
 - Parameterized by a key (K), **Key can be same/different for E and D**
 - Key is secret, but ciphers are considered public knowledge



Simple Encryption Scheme

Substitution Cipher: substituting one thing for another

❑ Monoalphabetic Cipher: substitute one letter for another

Original:

ABCDEFGHIJKLMNOPQRSTUVWXYZ

Substitute:

MNBVCXZASDFGHJKLPOIUYTREWQ

Example

Plaintext:

ALICE: I LIKE NETWORK SECURITY

Ciphertext:

MGBC: S GSFC JCURKOF ICBYOSUW

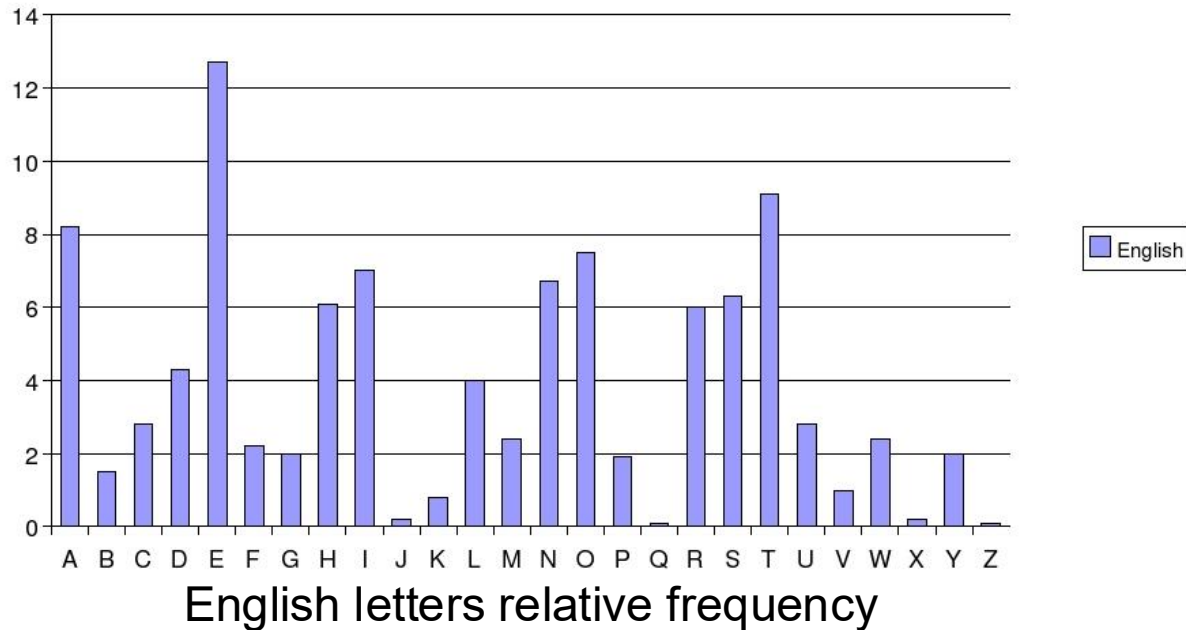


Encryption Key?

A mapping from set of 26 letters to set of 26 letters

Attacks: Monoalphabetic Cipher

- ❑ Easy to learn patterns
- ❑ Frequency analysis



A More Sophisticated Approach

- n substitution ciphers, $M_1, M_2, M_3, \dots, M_n$
- Examples: $n=4$ with cycling pattern of length 5
 - M_1, M_3, M_2, M_4, M_3
 - For each new plaintext symbol, use a subsequent substitution pattern in cyclic
 - **MAGIC**: **M** from M_1 , **A** from M_3 , **G** from M_2 , **I** from M_4 , **C** from M_3



Encryption Key?

n substitution ciphers, and cyclic pattern

Cryptanalysis: Breaking Encryption Schemes

❑ Cipher-text only attack

- Trudy has ciphertext, she can analyse
- Two approaches:
 - Brute force
 - Statistical analysis

❑ Chosen-plaintext attack

- Trudy can get ciphertext for a chosen plaintext

❑ Chosen-ciphertext attack

- Trudy can get plaintext for a chosen ciphertext

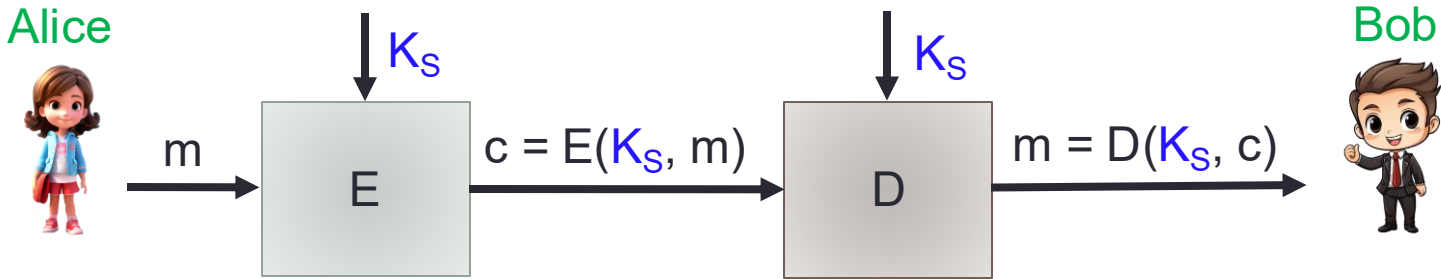
❑ Known-plaintext attack

- Trudy has plaintext corresponding to some cipher text
 - E.g., in monoalphabetic cipher, Trudy determines pairing for ALICE
 - Enigma code breaker in Word War II

❑ Side-channel attack

- E.g., Power analysis, Timing attack

Symmetric Key Cryptography



- Alice and Bob share same symmetric key (K_s)
 - E.g., In monoalphabetic substitution cipher, the key is substitution pattern known by both party
- Other Algorithms: DES, 3-DES, AES

Q: How do Alice and Bob agree on key value?

A: Diffie-Hellman Key exchange Protocol (part of Public Key Crypto)

B: Offline method

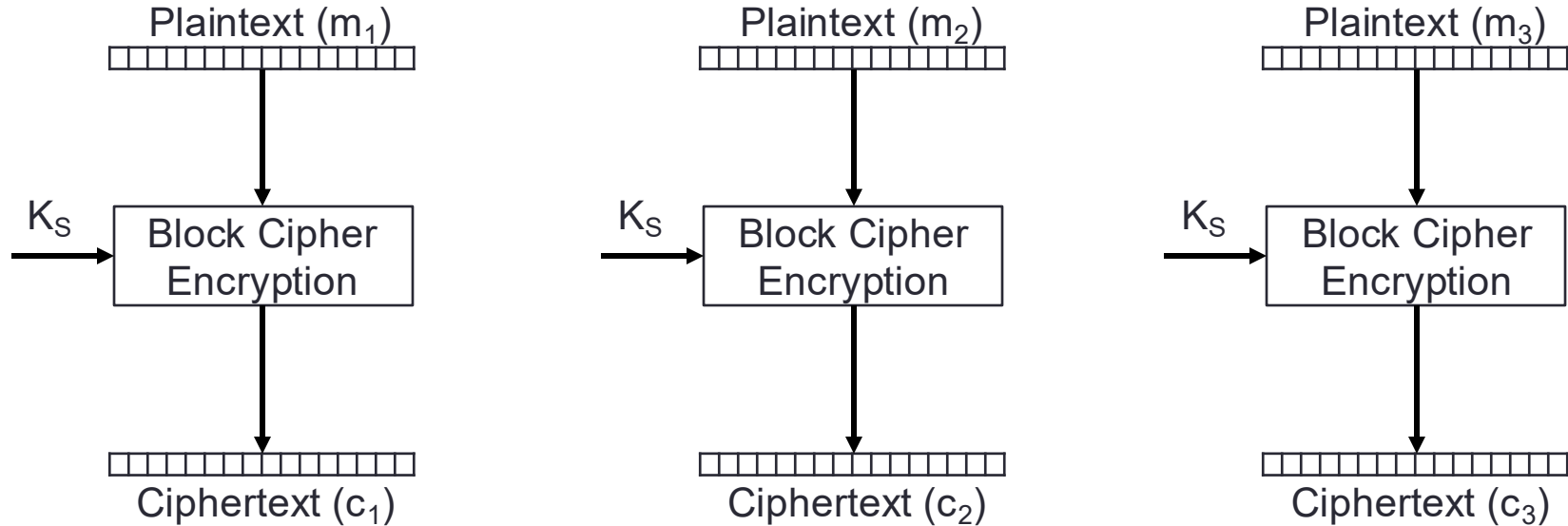
AES: Advanced Encryption Standard

❑ Symmetric Key NIST standard

- Replaced Data Encryption Standard (DES) which used 56-bit keys and processes data in 64-bit blocks (Block Cipher)
- 6 times faster than DES
- Processes data in 128-bit blocks
- Can operate on 128, 192, or 256 bit keys and took 10, 12 or 14 rounds respectively for encryption/decryption
- Operates on 2 mode ECB and CBC

❑ Brute force decryption (trying each key) takes one second for DES, while it takes 149 trillion years for AES-256

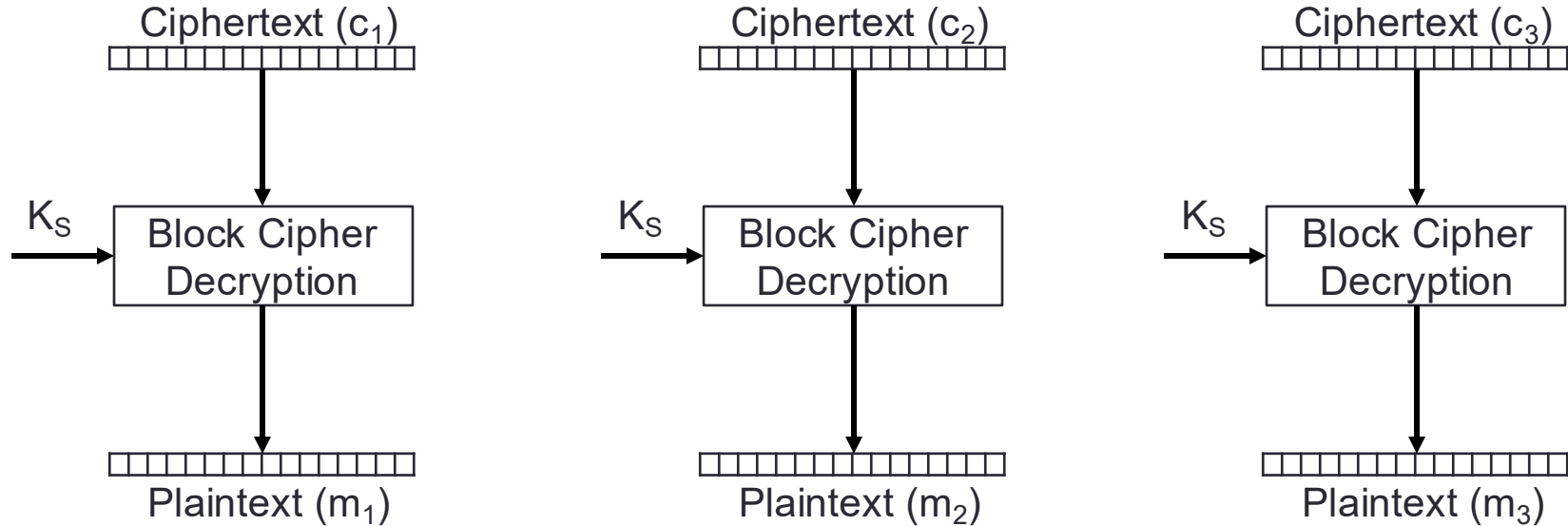
AES: Electronic Codebook (ECB) Mode



ECB Mode Encryption

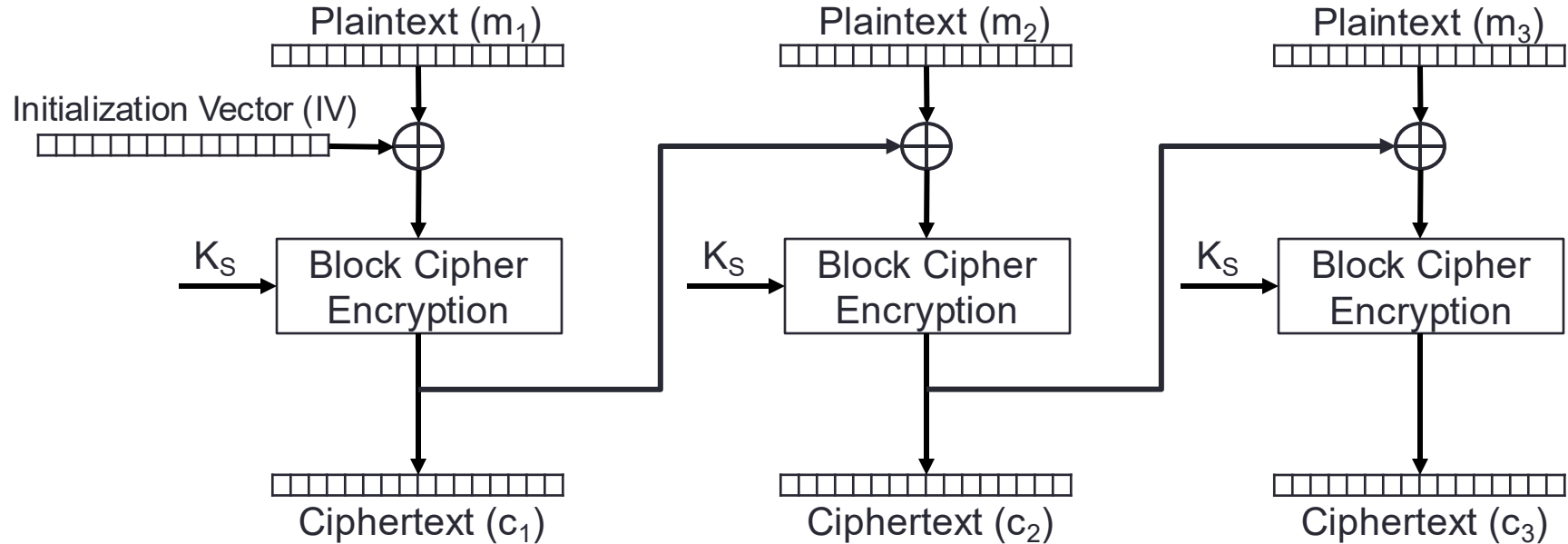
ECB encrypts identical plaintext blocks into identical ciphertext blocks

AES: Electronic Codebook (ECB) Mode...



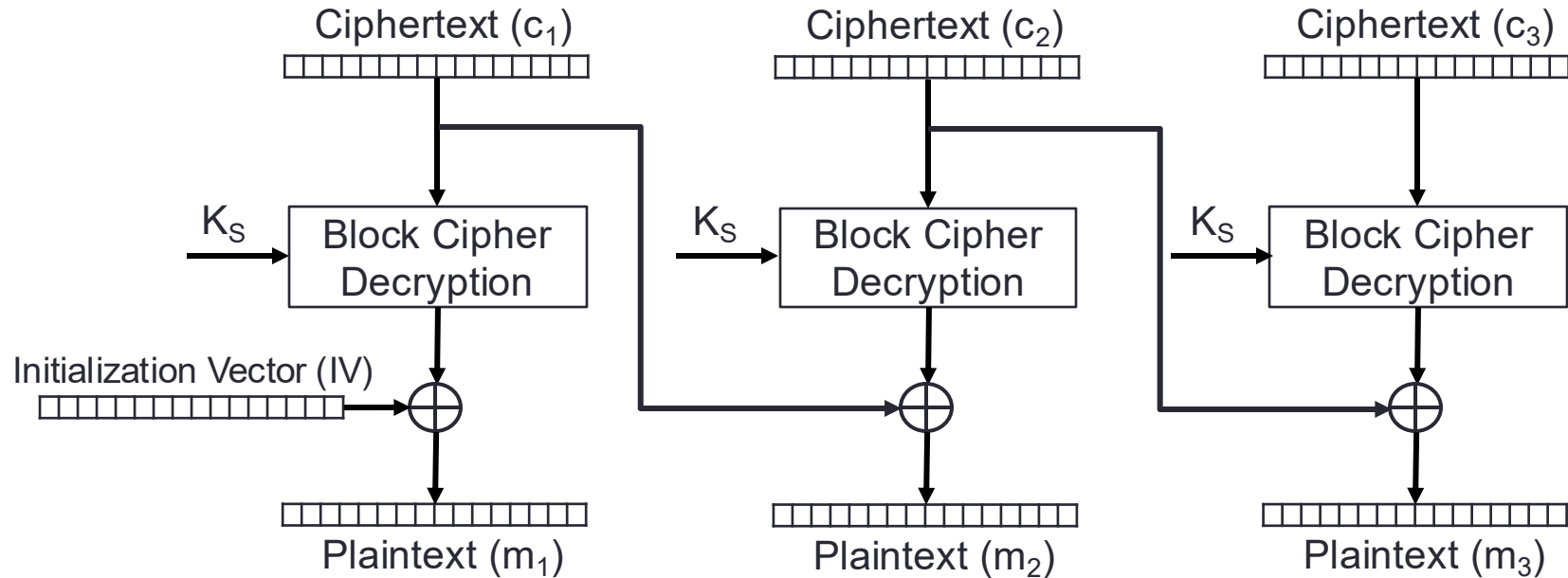
ECB Mode Decryption

AES: Cipher block Chaining (CBC) Mode



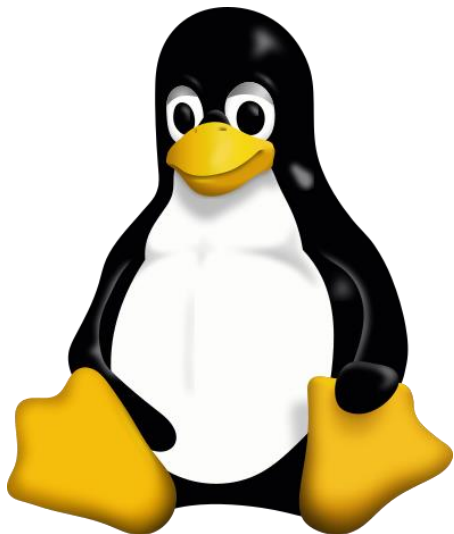
CBC Mode Encryption

AES: Cipher block Chaining (CBC) Mode...

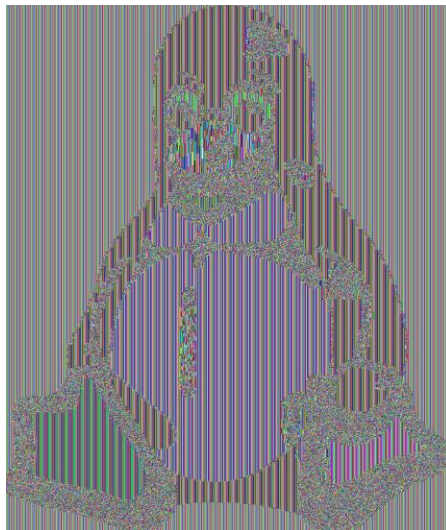


CBC Mode Decryption

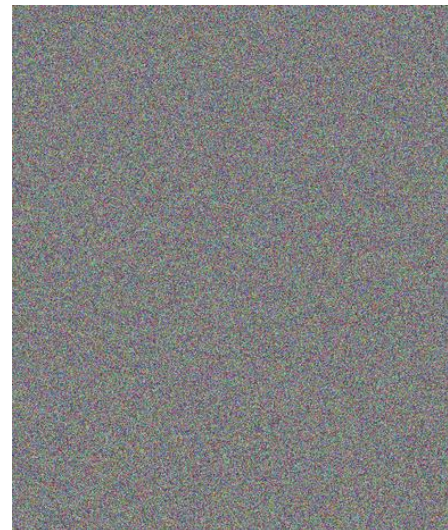
AES on Pictures



Original Image



Encryption using ECB mode



Encryption other than ECB like CBC results in pseudo-randomness

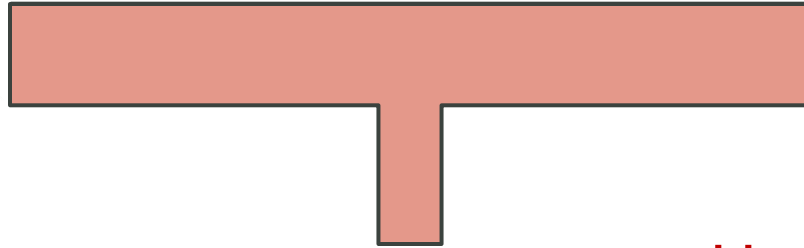
Public Key Cryptography

- ❑ Symmetric key crypto required sender and receiver know the shared secret key
 - How to agree on key in first place (particularly if never “met”)?
- ❑ Public key Crypto
 - Provides a protocol to exchange key (Diffie-Hellman, 1976)
 - Even work on a different approach, when comes to encryption/decryption (RSA , 1978)
 - Sender, receiver do **not** share secret key
 - **Public** encryption key is known to all
 - **Private** decryption key known only to receiver

How to Share Secret Key

- ❑ Symmetric key crypto requires sender and receiver know the shared key
 - How to share the key between two communicating party

Alice



Bob



Trudy



Use of Public Key
Cryptography

Key-Exchange: Diffie-Hellman Protocol

❑ First practical key exchange protocol to establish shared secret key between two communication party (**assuming only eavesdropping attack**)

❑ How it works?

➤ Fix a large prime **$p=23$** , an integer **$g=5$** in $\{1,2,\dots,p\}$, can be publicly known

Alice



Choose random **$a=4$** in $\{1,\dots,p-1\}$

compute **$A = g^a \pmod p$**
 $= 5^4 \pmod{23} = 4$

Combine **B** with **a**
 $K_S = B^a \pmod p$
 $= 10^4 \pmod{23} = 18$

$B=10$

$A=4$

Alice and Bob has same
secret key **$K_S = 10$**

Bob

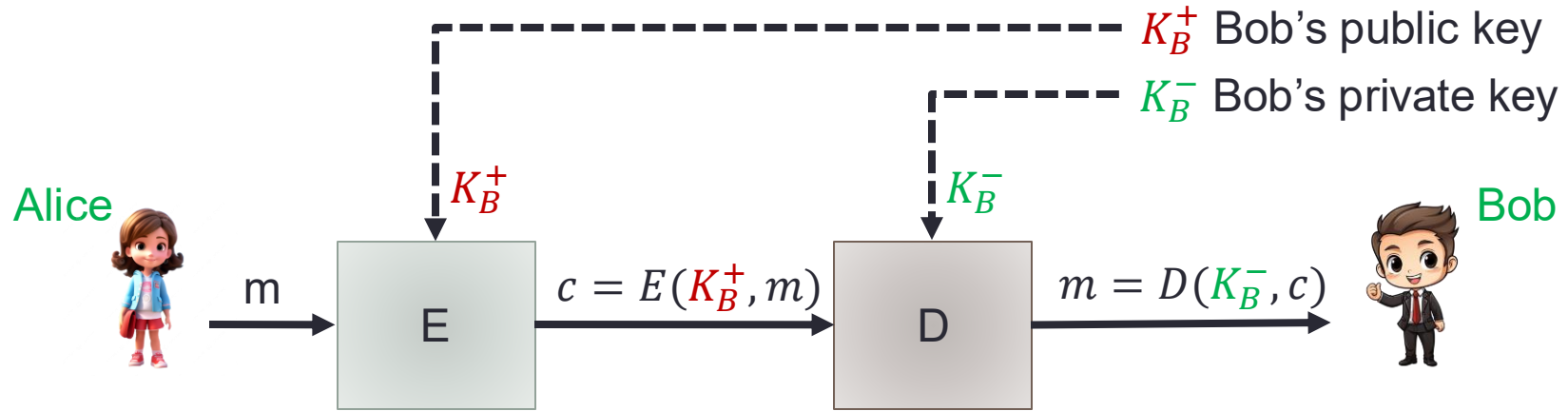


Choose random **$b=3$** in $\{1,\dots,p-1\}$

compute **$B = g^b \pmod p$**
 $= 5^3 \pmod{23} = 10$

Combine **A** with **b**
 $K_S = A^b \pmod p$
 $= 4^3 \pmod{23} = 18$

RSA: Public Key Encryption



Requirements:

- Need a key pair generation algorithm G that can generate K_B^+ and K_B^- such that,
 - $D(K_B^-, E(K_B^+, m))$
- Given public key K_B^+ , it should be impossible to compute private key K_B^-

RSA: Creating Public/Private Key Pair

STEP-1: Choose two large prime number p , q (e.g., 1024 bits each, or even more larger)

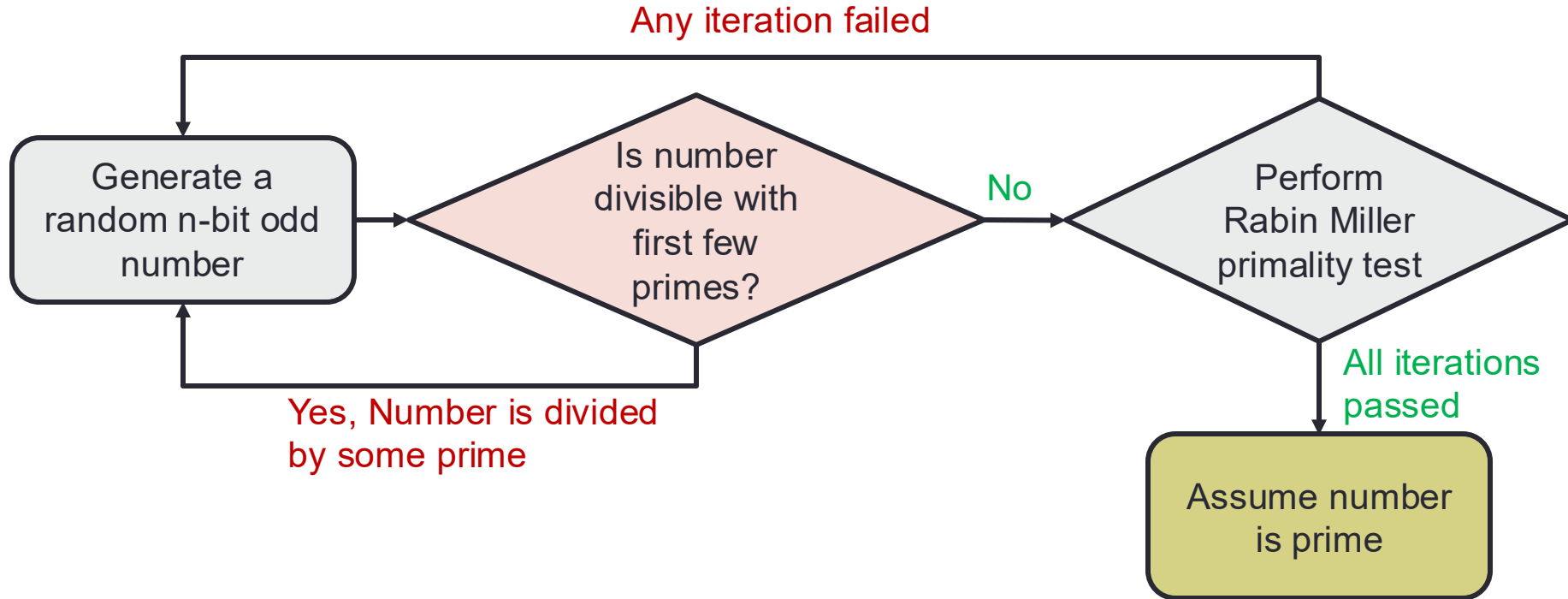
STEP-2: compute $n = pq$, $z = (p - 1)(q - 1)$

STEP-3: choose e (with $e < n$) that has no common factors with z (e , z are “relatively prime”)

STEP-4: choose d such that $(ed - 1)$ is exactly divisible by z . (in other words: $ed \pmod{z} = 1$)

STEP-5: public key is $\underbrace{(n, e)}_{K_B^+}$ and private key is $\underbrace{(n, d)}_{K_B^-}$

Generation of Large Primes



RSA: Encryption, Decryption in Action

Given (n, e) and (n, d) as computed previously

□ To encrypt plaintext message m ($< n$), compute

$$c = m^e \bmod n$$

□ To decrypt received ciphertext c , compute

$$m = c^d \bmod n$$

Magic is

$$m = \underbrace{(m^e \bmod n)}_c^d \bmod n$$

RSA Example

Bob chooses $p = 5$, $q = 7$, then $n = 35$, $z = 24$

$e=5$ (so e , z relatively prime)

$d=29$ (so $ed-1$ exactly divisible by z)

encrypting 8-bit message

encrypt: $\underbrace{m}_{12} \quad \underbrace{m^e}_{248832} \quad \underbrace{c = m^e \bmod n}_{17}$

decrypt: $\underbrace{c}_{17} \quad \underbrace{c^d}_{481968572106750915091411825223071697} \quad \underbrace{m = c^d \bmod n}_{12}$

Why is RSA Secure?

- ❑ Suppose you know Bob's public key (n, e) . How hard is it to determine d in his private key (n, d) ?
- ❑ Essentially need to find factors of n without knowing the two factors p and q
 - $n = p \times q$ is easy to calculate but hard to reverse
 - fact: factoring a large number is really hard
 - So, uses relatively large keys (1024 bits or more) and relies on the high computational cost of factoring large numbers

RSA in Practice: Session Keys

- ❑ Exponentiation in RSA is computationally intensive
 - But encryption is 10-30 times faster than decryption
- ❑ DES (symmetric key crypto) is at least 100 times faster than RSA (public key crypto)
- ❑ Use public key crypto to securely exchange a second key – symmetric session key – for encrypting application data faster

Session key K_S

- ❑ Alice and Bob use RSA to exchange a symmetric key K_S
- ❑ Once both have K_S , they use symmetric key cryptography (e.g, AES)

Private Vs. Public Key Cryptography

Symmetric Key Crypto

- ❑ Single (secret) key for encryption/decryption
- ❑ Communication among n people, it needs $n * (n - 1)/2$ secret keys
 - Each person has to maintain $n - 1$ keys secretly
- ❑ Encryption is very fast
 - Large data transfer
- ❑ Provide only confidentiality, authenticity, and integrity

Public Key Crypto

- ❑ Public key for encryption and private for decryption
- ❑ Needs only n number of <public, private> key pairs
 - Each person has to maintain only one private key secretly
- ❑ Encryption process is slow
 - Small data transfer like session keys
- ❑ Also provides non-repudiation